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FIN302

Finance and Markets

Lecture 11: Option Pricing

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Lecture 11-Option Pricing

Questions and Learning Outcomes

- How to pricing options and what is the pricing mechanism?
 - Binomial Tree
 - Random walk and binomial tree stock prices
 - No-arbitrage Pricing
 - Risk Neutral Pricing
 - Multiple-step Trees and American Options
 - Black-Scholes Option Price
 - Log-normal stock price
 - Black-Scholars Formula
 - Dividends and American Options



Lecture 11-Option Pricing

- Readings:
 - Hull (2022), Chapters 13, 15
- Tutorial 11 Problems :
 - 13.11, 12, 14, 16, 17, 22
 - 15.13, 21

Option Pricing: Binomial Tree

➤ *Binomial tree:*

- a diagram representing **all possible paths** followed by the asset price **in discrete-time**
- **assumption:** the asset price follows a *random walk*
 - at each time step price either moves up or down.
 - specified by
 - the **sizes** of the up and down movements
 - the **probability** of an up-movement

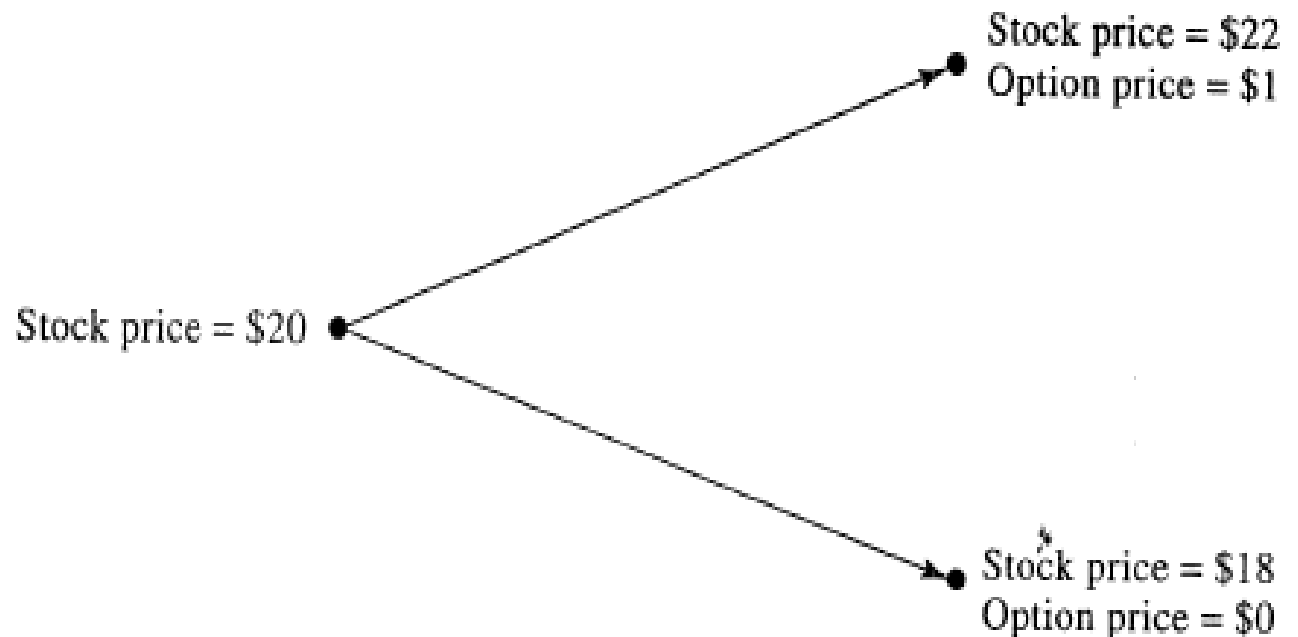
➤ *Binomial tree option pricing:*

- arbitrage argument
- *risk-neutral valuation*

One-step Binomial Tree

Example: A non-dividend-paying stock is currently trading for \$20, and it is known that the stock price after **three months** will be either \$22 or \$18. The risk-free interest rate is 12% per annum, with continuous compounding. How to value a three-month European call on the stock, with a strike price of \$21.

*Stock price and
Option payoff*



One-step Binomial Tree: No Arbitrage

➤ Example (continued):

- Consider a portfolio: **long 0.25 shares of stock** and **short a call** option.
 - The value of the portfolio:
 - At $t=0$: $20 \times 0.25 - f = 5 - f$
 - The payoff at T :
 - $22 \times 0.25 - 1 = 4.5$ if the stock price moves up;
 - $18 \times 0.25 = 4.5$ if the stock price moves down
- The terminal portfolio value is \$4.5, irrespective of whether the stock price moves up or down. There is no risk.
- Therefore, $5 - f = 4.5 e^{-0.12 \frac{3}{12}} = 4.367$, or $f \approx \$0.633$.

One-step Binomial Tree: No Arbitrage

- **Example** (continued): In general,
 - Consider a portfolio: **long Δ shares of stock** and **short a call** option.
 - The value (payoff) of this portfolio will be
 - $22\Delta - 1$ if the stock price moves up, and
 - 18Δ if the stock price moves down.
 - To make the portfolio payoff riskless we choose Δ so that
$$22\Delta - 1 = 18\Delta \quad \Rightarrow \quad \Delta = 0.25$$
 - The terminal portfolio value is then
$$22\Delta - 1 = 22 \times 0.25 - 1 = \$4.50 = 18 \times 0.25 = 18\Delta,$$
irrespective of whether the stock price moves up or down.

One-step Binomial Tree: No Arbitrage

➤ Example (continued):

- In the absence of arbitrage opportunities, the initial value of the portfolio must therefore be

$$4.5 e^{-0.12 \frac{3}{12}} = 4.367$$

- The initial value of the portfolio can also be written as

$$20\Delta - f = 20 \times 0.25 - f = 5 - f.$$

- From this,

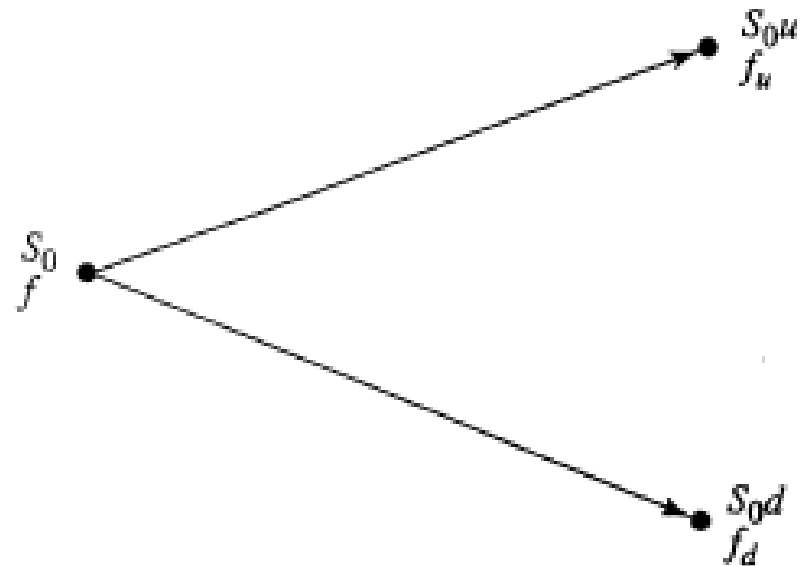
$$5 - f \approx 4.367 \quad \Rightarrow \quad f \approx \$0.633.$$

- **Note:** If the option does not trade at \$0.633, then an arbitrageur could generate a risk-free profit without any initial investment.

One-step Binomial Tree: No Arbitrage

- **One-step Binomial Option Pricing**-consider an option on a stock:
 - the current price is S_0 ,
 - the stock price S_T at time T will be either S_0u or S_0d , where $d < 1 < u$.
- Let f be the current value of an option on the stock that matures at time T , with the following payoff:

$$\text{payoff}(S_T) = \begin{cases} f_u & \text{if } S_T = S_0u \\ f_d & \text{if } S_T = S_0d. \end{cases}$$



No Arbitrage Argument

- No arbitrage portfolio:

- *long Δ shares of stock* and *short a call option*.

- The value of this portfolio at time T

$$\text{value}(S_T) = \begin{cases} S_0 u \Delta - f_u & \text{if } S_T = S_0 u \\ S_0 d \Delta - f_d & \text{if } S_T = S_0 d. \end{cases}$$

- To make the portfolio riskless, we must choose Δ so that

$$S_0 u \Delta - f_u = S_0 d \Delta - f_d \quad \Rightarrow \quad \Delta = \frac{f_u - f_d}{S_0 u - S_0 d}$$

- If the market is free of arbitrage opportunities, then the initial portfolio value must be

$$(S_0 u \Delta - f_u)e^{-rT} = (S_0 d \Delta - f_d)e^{-rT}$$



No Arbitrage Argument

- If we express the **initial value of the portfolio** as $S_0\Delta - f$, we must have

$$S_0\Delta - f = (S_0u\Delta - f_u)e^{-rT} \rightarrow f = S_0\Delta(1 - ue^{-rT}) + f_ue^{-rT}$$

- After substituting Δ into the above expression, we obtain

$$f = e^{-rT}[p f_u + (1 - p)f_d], \quad p = \frac{e^{rT} - d}{u - d}$$

- This expression for the option price *does not depend on*
 - *the expected stock price*
 - the probabilities of the stock price going up and down



Risk Neutral Probability

➤ Two Observations:

- If we treat p as the probability of stock price going up, we can view

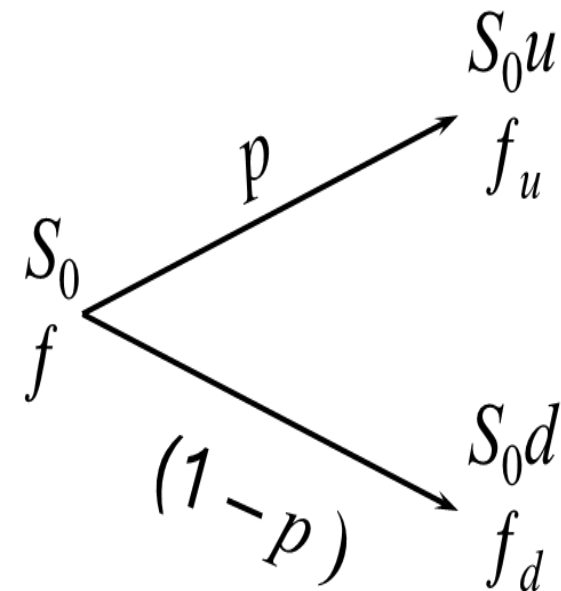
$$p f_u + (1 - p) f_d$$

as the **expected payoff of the option**; i.e.

$$E^*(f_T) = p f_u + (1 - p) f_d$$

- The expected payoff of the option generates a rate of return at the risk free rate

$$f_0 = e^{-rT} E^*(f_T)$$



Risk Neutral Valuation

- Therefore, we introduce the **fictional concept** of a ***risk-neutral world***.
 - The investors in a risk-neutral world are indifferent to risk, and care only about expected returns.
 - Consequently, all assets must exhibit the same expected returns in **the equilibrium** state of a risk-neutral world, equal to the **risk-free interest rate r** .
- **Risk-neutral probability:**
 - The probability p is called risk-neutral probability; meaning the stock price goes up with p and down with $1 - p$
 - The expectation **$E^*(f_T) = p f_u + (1 - p) f_d$** is ***the expected payoff under the risk-neutral probability measure***

Risk Neutral Probability

- **Note:** Risk-neutral valuation produces exactly the same option prices as obtained from arbitrage arguments.
- This is a special case of the so-called *Fundamental Theorem of Asset Pricing*:
 - *A financial market is free of arbitrage opportunities if and only if there exists an equivalent risk-neutral probability measure.*

Risk Neutral Probability

- In a risk-neutral world we must have

$$S_0 = e^{-rT} E^*(S_T),$$

where $E^*(\cdot)$ denotes expectation with respect to the **risk-neutral probability measure**.

- If we denote the *risk-neutral probability* of an up-movement in the stock price by p , then

$$E^*(S_T) = pS_0u + (1-p)S_0d = S_0e^{rT}, \quad p = \frac{e^{rT} - d}{u - d}$$

- Therefore, we must have

$$f_0 = e^{-rT} E^*(f_T) = e^{-rT} [p f_u + (1-p)f_d]$$



Risk Neutral Probability

➤ **Example:** We apply risk-neutral valuation to the call option

- Let p be the risk-neutral probability of an up-movement, then the initial asset price is the discounted expected value of the asset at T (under the risk-neutral probability measure)

$$S_0 = e^{-rT} E^*(S_T) = e^{-rT} [pS_u + (1 - p)S_d]$$

$$20 = e^{-0.12 \times \frac{3}{12}} [22p + (1 - p)18]$$

$$22p + (1 - p)18 = 20e^{0.12 \times \frac{3}{12}} \Rightarrow p = 0.6523$$

- The value of the option is the discounted expected payoff of the option (under the risk-neutral probability measure)

$$f_0 = e^{-rT} E^*(f_T) \Rightarrow f_0 = e^{-rT} [pf_u + (1 - p)f_d]$$

$$f_0 \approx e^{-0.12 \times \frac{3}{12}} \times [0.6523 \times 1 + (1 - 0.6523) \times 0] \approx \$0.633$$



Multiple-Step Tree

- We can extend the one-step tree to two-step tree or above:
 - T : the option maturity
 - N : the number of time-steps in the asset price tree
 - $\Delta t = \frac{T}{N}$: the length of a time step
- The expected price of any asset after one time-step, under the risk-neutral probability measure, is thus

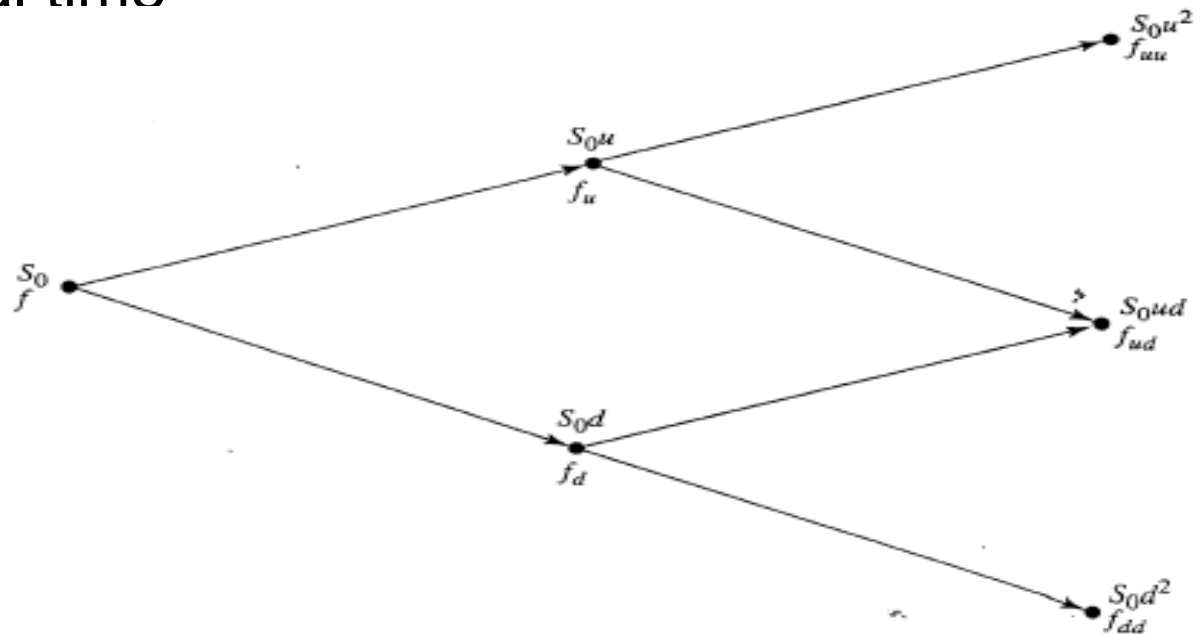
$$S_t = e^{-r\Delta t} E^*(S_{t+\Delta t}), \quad f_t = e^{-r\Delta t} E^*(f_{t+\Delta t})$$

- From this we obtain

$$pS_0u + (1-p)S_0d = S_0e^{r\Delta t} \Rightarrow p = \frac{e^{r\Delta t} - d}{u - d},$$
$$f_0 = e^{-r\Delta t}(pf_u + (1-p)f_d).$$

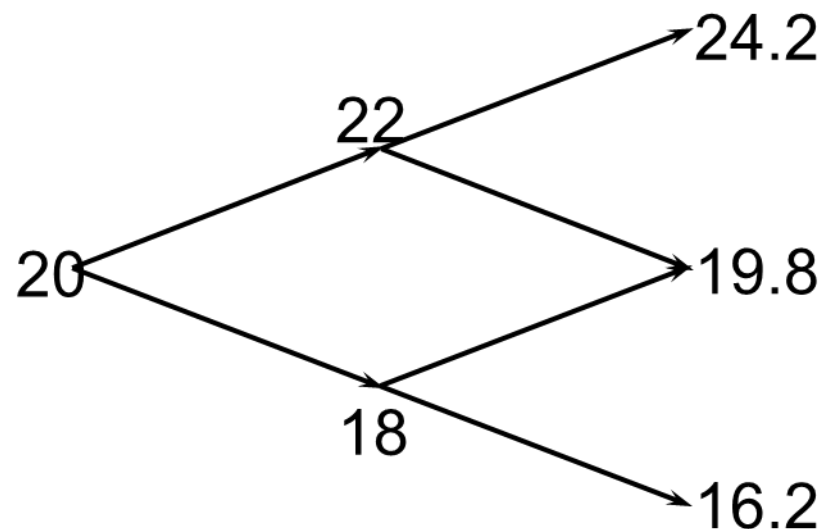
Two-Period Binomial Tree---Europe Options

- **Two-period binomial tree---Risk-neutral valuation:**
- First calculate the option payoffs f_{uu} , $f_{ud} = f_{du}$ and f_{dd} at the terminal nodes.
 - Then use risk-neutral valuation to determine the option prices f_u and f_d at the two intermediate nodes.
 - Finally, use risk-neutral valuation to determine the option price f at the initial time



Two-Period Binomial Tree---European Options

- **Example:** Consider a *European call*, with a strike price of \$21, on a non-dividend-paying stock whose current price is \$20.
- The price of the stock is assumed to follow the two-step binomial tree over the next six months (so each time-step is three months).
 - The continuously compounded risk-free interest rate is 12% per annum, for all maturities up to six months.



Two-Period Binomial Tree---European Options

➤ Example:

- The up- and down-factors for the tree are

$$u = \frac{22}{20} = 1.1, \quad d = \frac{18}{20} = 0.9$$

- The risk-neutral probability of an up-movement across any time step is

$$p = \frac{e^{r\Delta t} - d}{u - d} = \frac{e^{0.12 \frac{3}{12}} - 0.9}{1.1 - 0.9} = 0.6523$$

where $\Delta t = 3/12 = 0.25$ years is the length of a time-step.

Two-Period Binomial Tree---European Options

➤ Example (continued):

- First we determine the payoffs of the option at the terminal nodes (see nodes D–F in the Figure):

$$f_{uu} = 3.2, \quad f_{ud} = f_{du} = 0 \quad \text{and} \quad f_{dd} = 0.$$

- Then we use risk-neutral valuation to determine the option price after one up-move (see node B in the Figure):

$$\begin{aligned} f_u &= e^{-r\Delta t} (pf_{uu} + (1-p)f_{ud}) \\ &\approx e^{-0.12 \times \frac{3}{12}} (0.6523 \times 3.2 + (1 - 0.6523) \times 0) \approx \$2.0257 \end{aligned}$$

- Next, we use risk-neutral valuation to determine the option price after one down-move

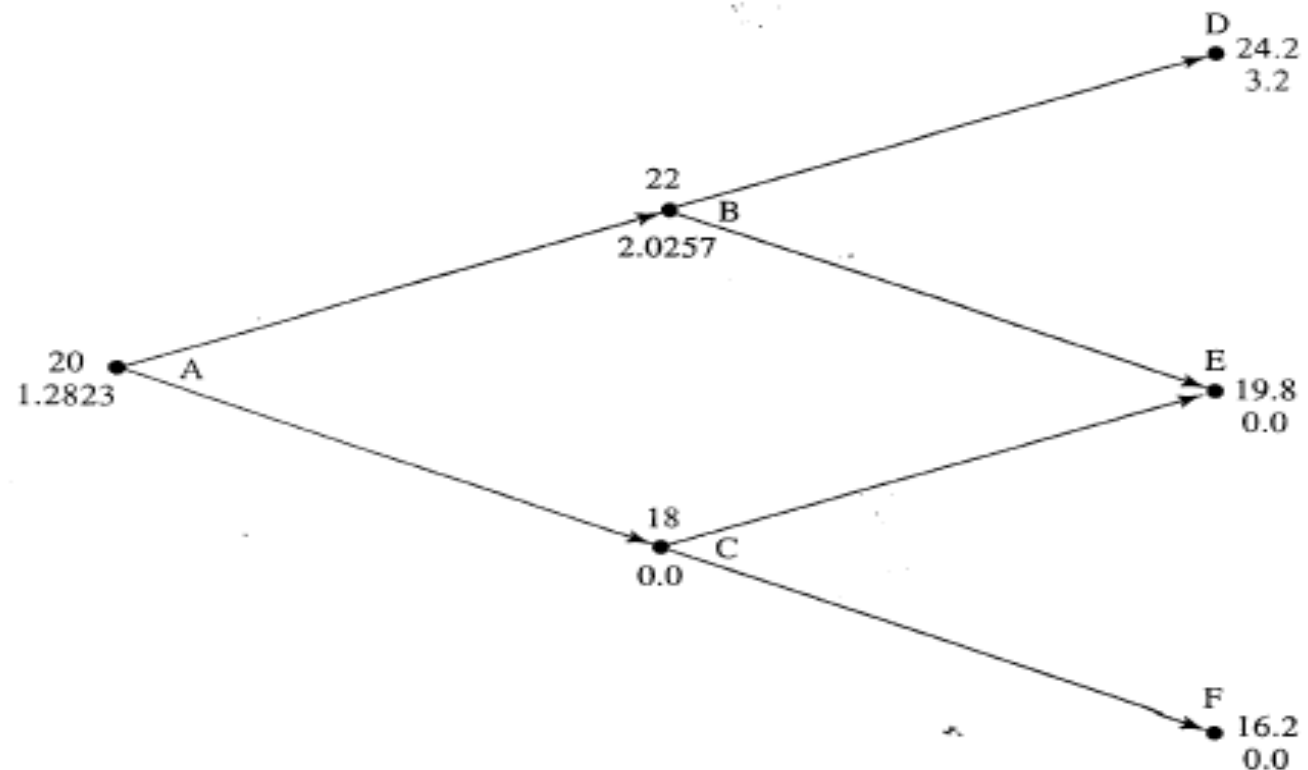
$$\begin{aligned} f_d &= e^{-r\Delta t} (pf_{du} + (1-p)f_{dd}) \\ &= e^{-0.12 \times \frac{3}{12}} (0.6523 \times 0 + (1 - 0.6523) \times 0) = \$0.00 \end{aligned}$$

Two-Period Binomial Tree---European Options

➤ Example (continued):

- Finally, we use risk-neutral valuation to determine the initial option price:

$$f = e^{-r\Delta t}(pf_u + (1-p)f_d)$$
$$\approx e^{-0.12 \times \frac{3}{12}}(0.6523 \times 2.0257 + (1 - 0.6523) \times 0) \approx \$1.2823$$



Two-Period Binomial Tree---American Option

- To determine the price of an **American option** with a two-step binomial tree, we modify the risk-neutral valuation procedure as follows:
 - First calculate the option payoffs f_{uu} , $f_{ud} = f_{du}$ and f_{dd} at the terminal nodes.
 - Then use risk-neutral valuation to determine the option prices f_u and f_d at the two intermediate nodes.
 - Next, **replace f_u and f_d with the maxima of their original values and the payoffs that result from early exercise at the intermediate nodes.**
- **Note:** With **non-dividend-paying stocks**, the price of an **American call** is always the same as the price of the equivalent European call

Two-Period Binomial Tree---American Option

Example: Consider an American put option with $K = 52$, $y = 5\%$, $T = 2$ (see the Figure).

- The payoffs at maturity are the same as before

$$f_{uu} = 0, \quad f_{ud} = f_{du} = 4, \quad f_{dd} = 20$$

- The option value at the upper intermediate node is

$$f_u = \max\{1.4147, 0\} = 1.4147$$

- The option price at the lower intermediate node is

$$f_d = \max\{9.4636, 12\} = 12$$

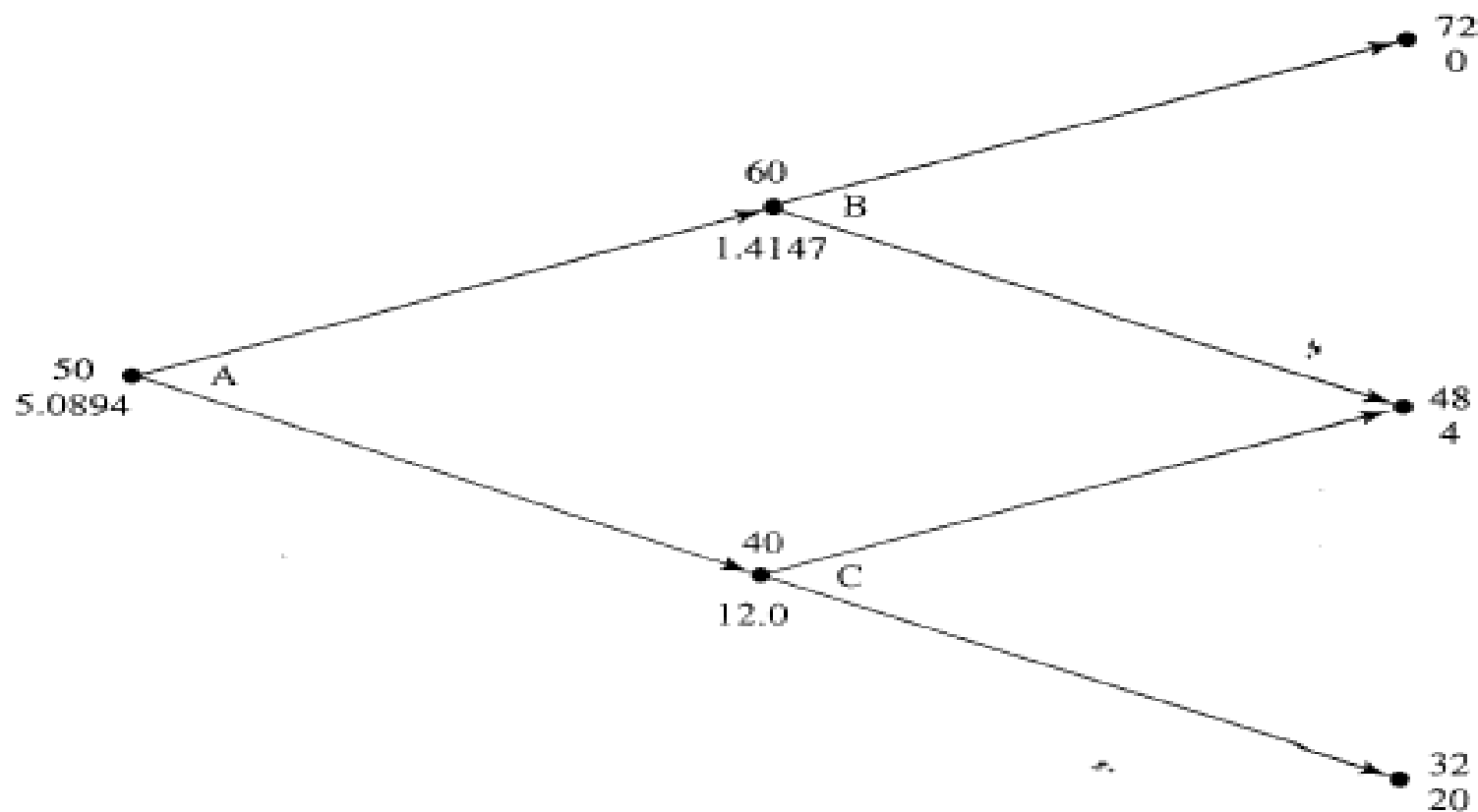
meaning **it is optimal to exercise the option immediately**

- The initial option price is therefore

$$\begin{aligned} f &= \max\left\{e^{-r\Delta t}(pf_u + (1-p)f_d), \max\{K - S_0, 0\}\right\} \\ &= \max\left\{e^{-0.05 \times 1}(0.6282 \times 1.4147 + (1 - 0.6282) \times 12), 2\right\} \\ &= \$5.0894. \end{aligned}$$

Two-Period Binomial Tree---American Option

Fig.5



Delta Hedge: Δ

- The **delta of an option** is the ratio of the change in the option price to the change in the price of the underlying asset,

$$\Delta = \frac{f_u - f_d}{S_0 u - S_0 d} = \frac{f_u - f_d}{S_0(u - d)}$$

- It is also the number of units of the underlying asset one should hold in order to **hedge** the option.
 - In other words, a portfolio consisting of a short position in the option and a long position in Δ units of the underlying asset will be riskless.
- The **delta is positive for a call, but negative for a put**
- **Example:** For the European call in the Figure, we have $\Delta = 0.25$, $\Delta_u = 0.7273$ and $\Delta_d = 0$ ---**dynamic hedging**
- **Example:** For the European put in the Figure, we have $\Delta = -0.4024$, $\Delta_u = -0.1667$ and $\Delta_d = -1$



Black-Scholes Option Pricing

- The binomial asset price model is a discrete-time model because trading may only occur at discrete times
- By contrast, the option pricing formulae obtained by Black-Scholes and Merton (**BSM**) for European contracts on a non-dividend-paying stock were based on the assumption that the price of the underlying asset was described by a particular continuous-time model, called *geometric Brownian motion*.
- BSM formula is based on the random walk assumption:
 - *Returns are normally distributed*
 - *Prices are lognormally distributed*

The Normal Distribution

➤ Notation:

- S_0 : The initial price of a non-dividend-paying stock
- T : Some fixed future date
- S_T : The (random) price of the stock at time T
- μ : The expected return on the stock per year
- σ : The volatility of the stock price per year

Risk-Neutral Valuation

- The risk preferences of investors do not affect option prices.
- The preference-independent nature of the Black-Scholes formula makes it consistent with the **risk-neutral valuation principle**.
- The current asset price satisfies

$$S_0 = e^{-rT} E(S_T)$$

- The current value of a **call option**

$$\begin{aligned} c &= e^{-rT} E(c_T) = e^{-rT} E(\max\{S_T - K, 0\}) \\ &= e^{-rT} E(S_T - K | S_T \geq K) \text{Prob}(S_T \geq K) \\ &= e^{-rT} E(S_T | S_T \geq K) \text{Prob}(S_T \geq K) \\ &\quad - K e^{-rT} \text{Prob}(S_T \geq K) \end{aligned}$$

$$c = S_0 N(d_1) - K e^{-rT} N(d_2)$$

- Therefore,

$$\begin{aligned} S_0 N(d_1) &= e^{-rT} E(S_T | S_T \geq K) \text{Prob}(S_T \geq K) \\ N(d_2) &= \text{Prob}(S_T \geq K) \end{aligned}$$



Black-Scholes Option Price

- Similarly, with the payoffs $\max\{K - S_T, 0\}$ for put, the prices of a **European put**, p , on a non-dividend paying asset, with strike price K and maturity T

$$\begin{aligned} p &= e^{-rT} E(p_T) = e^{-rT} E(\max\{K - S_T, 0\}) \\ &= e^{-rT} E(K - S_T | S_T \leq K) \text{Prob}(S_T \leq K) \\ &= K e^{-rT} \text{Prob}(S_T \leq K) \\ &\quad - e^{-rT} E(S_T | S_T \leq K) \text{Prob}(S_T \leq K) \\ p &= K e^{-rT} N(-d_2) - S_0 N(-d_1) \end{aligned}$$

- Where

$$\begin{aligned} S_0 N(-d_1) &= e^{-rT} E(S_T | S_T \leq K) \text{Prob}(S_T \leq K) \\ N(-d_2) &= \text{Prob}(S_T \leq K) \end{aligned}$$

$$d_1 = \frac{\ln \frac{S_0}{K} + \left(r + \frac{\sigma^2}{2}\right) T}{\sigma \sqrt{T}}, \quad d_2 = \frac{\ln \frac{S_0}{K} + \left(r - \frac{\sigma^2}{2}\right) T}{\sigma \sqrt{T}} = d_1 - \sigma \sqrt{T}$$

$N(d_1)$ and $N(d_2)$ can be interpreted as risk-neutral probabilities.

The Normal Distribution

- The function $N(x) := P(Z \leq x)$ is called the **cumulative distribution function (CDF)** of $Z \sim \phi(0,1)$
 - The **NORMSDIST** function in Excel can be used to compute standard normal cumulative probabilities.
 - Standard normal cumulative normal probabilities can also be estimated from the tables of the textbook.
- **Example:** Using linear interpolation and the table,

$$\begin{aligned} N(-0.4567) &\approx N(-0.45) - 0.67[N(-0.45) - N(-0.46)] \\ &= 0.3264 - 0.67(0.3264 - 0.3228) = 0.3240. \end{aligned}$$

Black-Scholes Option Price

Example: A non-dividend-paying stock is trading for \$42. Consider European options on the stock, strike at \$40 and maturing after six months. The continuously compounded risk-free interest rate is 10% per annum, and the stock price volatility is 20% per annum.

$$\begin{aligned}d_1 &= \frac{\ln \frac{S_0}{K} + (r + \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}} \\&= \frac{\ln \frac{42}{40} + (0.1 + \frac{1}{2} \times 0.2^2) \times 0.5}{0.2 \times \sqrt{0.5}} \approx 0.7693, \quad \text{and} \\d_2 &= \frac{\ln \frac{S_0}{K} + (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}} \\&= \frac{\ln \frac{42}{40} + (0.1 - \frac{1}{2} \times 0.2^2) \times 0.5}{0.2 \times \sqrt{0.5}} \approx 0.6278.\end{aligned}$$

Black-Scholes Option Price

Example: Using the NORMSDIST function in Excel, we obtain (verify that the same values are obtained by using linear interpolation and the tables)

$$\begin{aligned}N(d_1) &\approx N(0.7693) \approx 0.7791; \\N(-d_1) &\approx N(-0.7693) \approx 0.2209; \\N(d_2) &\approx N(0.6278) \approx 0.7349; \\N(-d_2) &\approx N(-0.6278) \approx 0.2651.\end{aligned}$$

The Black-Scholes prices of the two options are therefore

$$\begin{aligned}c &= S_0 N(d_1) - Ke^{-rT} N(d_2) \\&\approx 42 \times 0.7791 - 40 \times e^{-0.1 \times 0.5} \times 0.7349 \approx \$4.76, \\p &= Ke^{-rT} N(-d_2) - S_0 N(-d_1) \\&\approx 40 \times e^{-0.1 \times 0.5} \times 0.2651 - 42 \times 0.2209 \approx \$0.81.\end{aligned}$$

Appendix

Dividend-Paying Asset

- To apply risk-neutral valuation to an option on an asset that pays income in the form of a yield q , we simply modify the expression for the risk-neutral probability p .
- In this case, the expected capital gain from the asset should therefore be $r - q$, under the risk-neutral probability measure.
- The expected price of the asset after one time step, under the risk-neutral probability measure, is thus

$$S_0 = e^{-(r-q)t} E^*(S_t)$$

- From this we obtain

$$pS_0u + (1 - p)S_0d = S_0e^{(r-q)\Delta t} \Rightarrow p = \frac{e^{(r-q)\Delta t} - d}{u - d}$$

Dividend-Paying Assets: index, currency, and futures

- For **options on stock indices**, q is the index dividend yield and

$$p = \frac{e^{(r-q)\Delta t} - d}{u - d}$$

- For **options on currencies**, $q = r_f$ is the foreign risk-free rate, r is the domestic risk-free rate and

$$p = \frac{e^{(r-r_f)\Delta t} - d}{u - d}$$

- For **options on futures**, $q = r$ the risk-free rate and

$$p = \frac{e^{(r-r)\Delta t} - d}{u - d} = \frac{1 - d}{u - d}$$

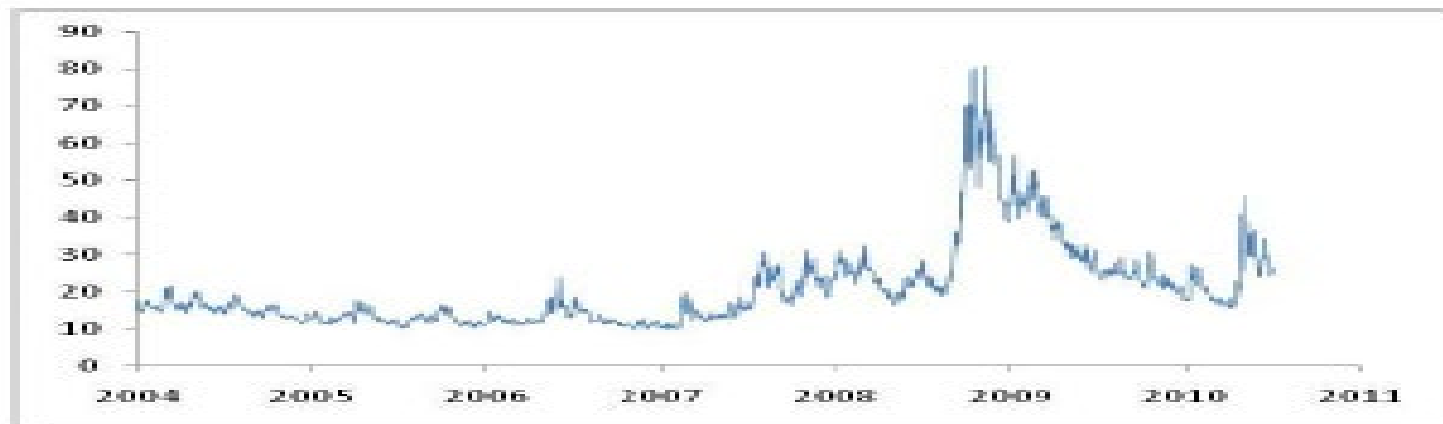
- As the options pays no dividends, the discounted expected payoff of the option under the risk-neutral probability measure is

$$f_0 = e^{-rT} E^*(f_T) = e^{-rT} [pf_u + (1 - p)f_d]$$



Implied Volatility

- The implied volatility of an option is the volatility for which the Black-Scholes price equals the market price.
- There is a one-to-one correspondence between prices and implied volatilities.
- Computing implied volatility thus requires market price for options, and a mean of inverting the Black-Scholes formula.
- Typical approaches to invert the Black-Scholes formula are
 - Trial and error,
 - Excel's Goalseek tool (Data/What-If Analysis/Goal Seek).



Dividends and American Options

- **Dividends:** European options on dividend-paying stocks are valued by substituting into the Black-Scholes formula, the stock price S_0 with the difference between the stock price and the present value of the dividends $S^* = S_0 - D$
- **American options**
 - An American call on a non-dividend-paying stock should never be exercised early.
 - An American call on a dividend-paying stock should only ever be exercised immediately prior to an ex-dividend date.
 - The price of the American call is estimated as the maximum of the following two European option prices (Black's Approximation)
 - The European call maturing at the same date as the American option,
 - The European call maturing just before the final ex-dividend date.

The Normal Distribution

➤ **Normal random variable:** $X \sim \phi(m, v)$ to indicate that X is a with mean m and variance $v > 0$.

▪ **Standard normal random variable:** $Z \sim \phi(0, 1)$

▪ Note that $X \sim \phi(m, v) \Rightarrow \frac{X-m}{\sqrt{v}} \sim \phi(0, 1)$

➤ The *probability density function (PDF)* of $Z \sim \phi(0, 1)$ is given by

$$N'(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

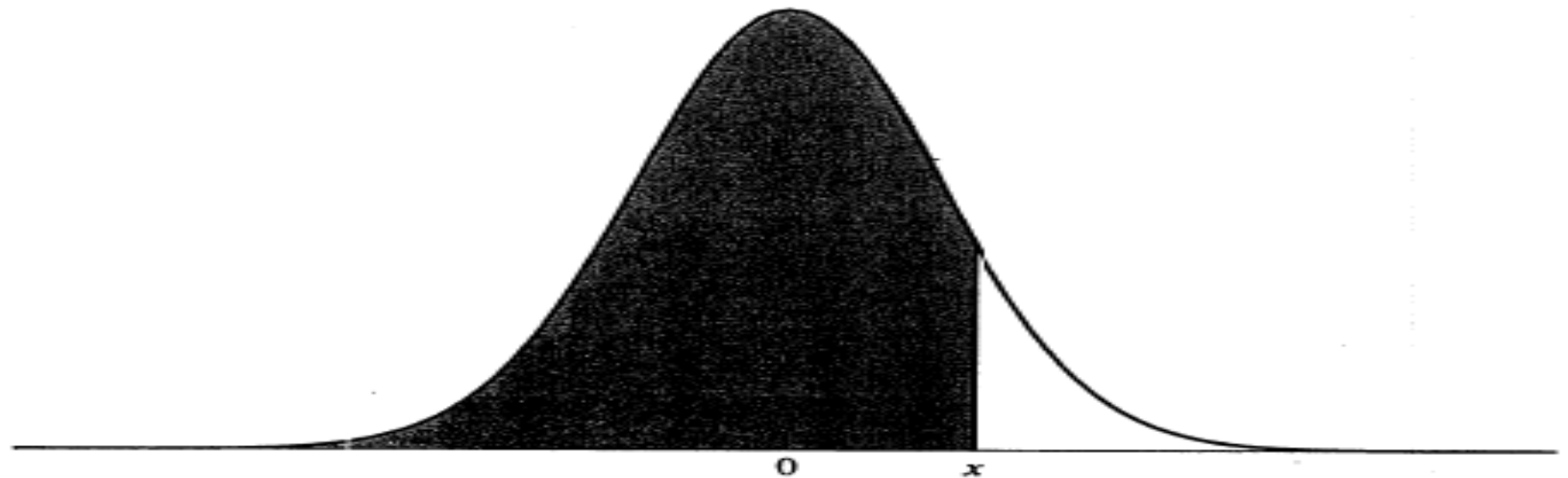
➤ The standard normal PDF is the characteristic bell-shaped curve

The Normal Distribution

- The following identity indicates that the CDF of a standard normal random variable measures the total area under its PDF to the left of its argument (see Figure):

$$N(x) = \int_{-\infty}^x N'(y) dy$$

- The standard normal PDF and CDF: The bell-shaped curve is $N'(x)$ and the shaded area underneath it is $N(x)$.



The Log-Normal Stock Price

- Under the random walk assumption, in a short period of time of length δt , the change in the stock price S_0 is normally distributed

$$\delta S \sim \phi(\mu S_0(\delta t), \sigma^2 S_0^2(\delta t))$$

- Thus ***the return on the stock is normally distributed***

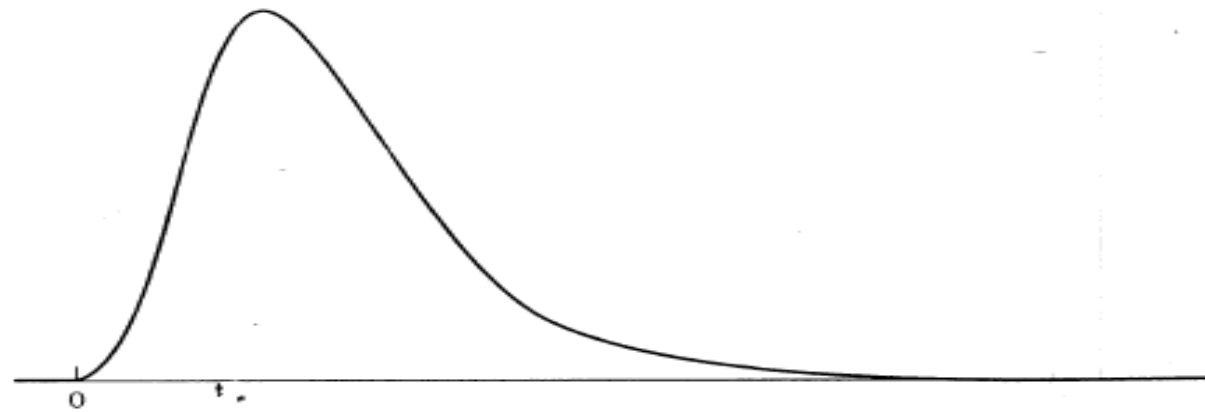
$$\frac{\delta S}{S_0} \sim \phi(\mu(\delta t), \sigma^2(\delta t))$$

- It follows that **the stock log-return is also normally distributed**
- Since the logarithm of S_T is normally distributed, **S_T is lognormally distributed** with

$$E(S_T) = S_0 e^{\mu T}, \quad \text{var}(S_T) = S_0^2 e^{2\mu T} (e^{\sigma^2 T} - 1)$$

The Log-Normal Stock Price

- Figure 2: The shape of a log-normal distribution



The Log-Normal Stock Price

- The realized continuously compounded return on the stock:

$$S_T = S e^{RT}, \quad R = \ln \frac{S_T}{S} \sim \phi \left(\left(\mu - \frac{\sigma^2}{2} \right) T, \sigma^2 T \right)$$

- It follows that the expected return (in a long period of time) expressed with continuous compounding

$$E(R) = \left(\mu - \frac{\sigma^2}{2} \right) T, \quad V(R) = \sigma^2 T$$

- **Example:** If a stock price is \$50 and its volatility is 25% per year what is the standard deviation of the price change in one day?

$$25\% \times \sqrt{\frac{1}{250}} = 1.57\% \text{ per day}$$



Black-Scholes Option Price: No-arbitrage

- The underlying concept of Black-Scholes Option Price:
 - The option price and the stock price depend on the same underlying source of uncertainty.
 - We can form a **delta hedging portfolio** consisting of the stock and option to eliminate the source of uncertainty.
 - The portfolio is instantaneously riskless and must earn the risk-free rate.
 - This is exactly the same approach as the binomial tree (CRR model) with

$$u = e^{\sigma\sqrt{\delta t}}, \quad d = e^{-\sigma\sqrt{\delta t}} = \frac{1}{u}, \quad p = \frac{e^{(r-q)(\delta t)} - d}{u - d}$$

just with very small time steps, $(\delta t) \rightarrow 0$.

Black-Scholes Option Price: No-arbitrage

➤ Let S be the stock price and $f = f(S, t)$ be the price of a call or put option of the stock

➤ Over a short-time interval $[t, t + (\delta t)]$,

$$\delta S = \mu S (\delta t) + \sigma S (\delta z)$$

$$\delta f = \left(\frac{\partial f}{\partial S} \mu S + \frac{\partial f}{\partial t} + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2 \right) (\delta t) + \frac{\partial f}{\partial S} \sigma S (\delta z)$$

➤ Notations---**Greek Letters**:

- **Delta:** $\Delta = \frac{\partial f}{\partial S}$
- **Theta:** $\Theta = \frac{\partial f}{\partial t}$
- **Gamma:** $\Gamma = \frac{\partial^2 f}{\partial S^2} = \frac{\partial \Delta}{\partial S}$

➤ Then
$$\delta f = \left(\Delta \mu S + \Theta + \frac{1}{2} \Gamma \sigma^2 S^2 \right) (\delta t) + \Delta \sigma S (\delta z)$$



Black-Scholes Option Price: No-arbitrage

➤ Consider a portfolio:

- Long shares: $\Delta = \frac{\partial f}{\partial S}$
- Short one option: $-f$

➤ Then the value of the portfolio: $\Pi = \Delta S - f$

➤ The change of the portfolio value

$$\delta \Pi = \Delta (\delta S) - (\delta f) = \left[-\Theta - \frac{1}{2} \Gamma \sigma^2 S^2 \right] (\delta t)$$

➤ Therefore,

$$\delta \Pi = r \Pi (\delta t); \quad -\Theta - \frac{1}{2} \Gamma \sigma^2 S^2 = r(\Delta S - f)$$

➤ Or

$$rf = \Theta + r \Delta S + \frac{1}{2} \Gamma \sigma^2 S^2$$



Risk-Neutral Valuation

- **Interpreting $N(d_1)$:** $S_0 N(d_1)$ gives the present value of the (risk neutral) expected stock price on the expiration date, given that the stock price is greater than K .
- The payoff of a call option is $S_T - K$; if $S_T > K$ and zero otherwise. Thus, from risk-neutral valuation

$$E^*(c_T) = E^*(S_T - K | S_T > K) \times P^*(S_T > K) + 0 \times P^*(S_T \leq K)$$

$$c_0 e^{rT} = E^*(S_T | S_T > K) P^*(S_T > K) - E^*(K | S_T > K) P^*(S_T > K)$$

- Given that $N(d_2) = P(S_T > K)$ and from the BS option pricing model, we have that

$$(S_0 N(d_1) - K e^{-rT} N(d_2)) e^{rT} = E^*(S_T | S_T > K) N(d_2) - K N(d_2)$$

$$S_0 e^{rT} N(d_1) = E^*(S_T | S_T > K) N(d_2)$$

$$S_0 N(d_1) = e^{-rT} E^*(S_T | S_T > K) N(d_2).$$

Interpreting $N(d_2)$

- $N(d_2)$ measures the risk-neutral probability that the option will be exercised, or equivalently, that the share price will exceed the strike price on expiration.
- In fact, under the BSM option pricing model

$$\ln \frac{S_T}{S_0} \sim \phi\left(\left(r - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right), \text{ or } Z_T = \frac{\ln \frac{S_T}{S_0} - \left(r - \frac{\sigma^2}{2}\right)T}{\sigma \sqrt{T}} \sim \phi(0, 1).$$

- Note that

- Thus
$$P^*(S_T \leq K) = P^*(Z_T \leq d_2^*) = N(d_2^*), \text{ where } d_2^* = \frac{\ln \frac{K}{S_0} - \left(r - \frac{\sigma^2}{2}\right)T}{\sigma \sqrt{T}}$$

$$P^*(S_T > K) = 1 - P^*(S_T \leq K) = 1 - N(d_2^*) = N(-d_2^*) = N(d_2).$$



Estimate Default Probability

- **Merton model:** the equity value E_0 of a firm:
- The value of the firm's equity at T: $E_T = \max(V_T - D, 0)$
 - V: firm value
 - D: debt repayment at T
 - The value of the firm's equity at 0

$$E_0 = V_0 N(d_1) - D e^{-rT} N(d_2)$$

$$d_1 = \frac{\ln\left(\frac{V_0}{D}\right) + \left(r + \frac{\sigma_V^2}{2}\right)T}{\sigma_V \sqrt{T}}, \quad d_2 = d_1 - \sigma_V \sqrt{T}$$

- The default probability:

$$\text{Prob}(V_T < D) = 1 - \text{Prob}(V_T \geq D) = 1 - N(d_2) = N(-d_2)$$